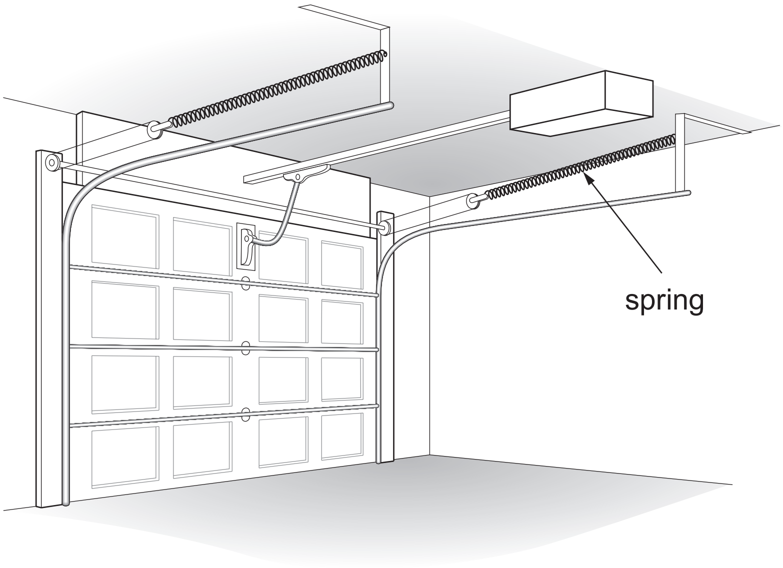
*Answer **all** the questions in the spaces provided.*

1. Springs can be used in garage doors. When the door shuts the springs are stretched and put under tension. When the door opens the spring returns to its original length making it easier to raise.



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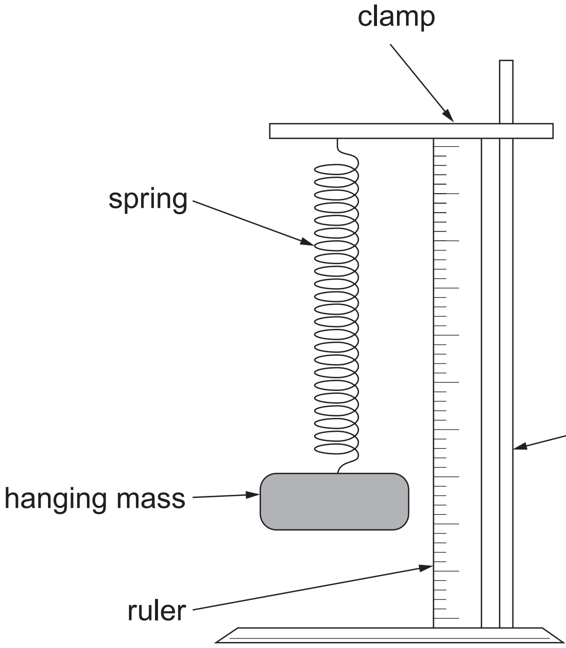
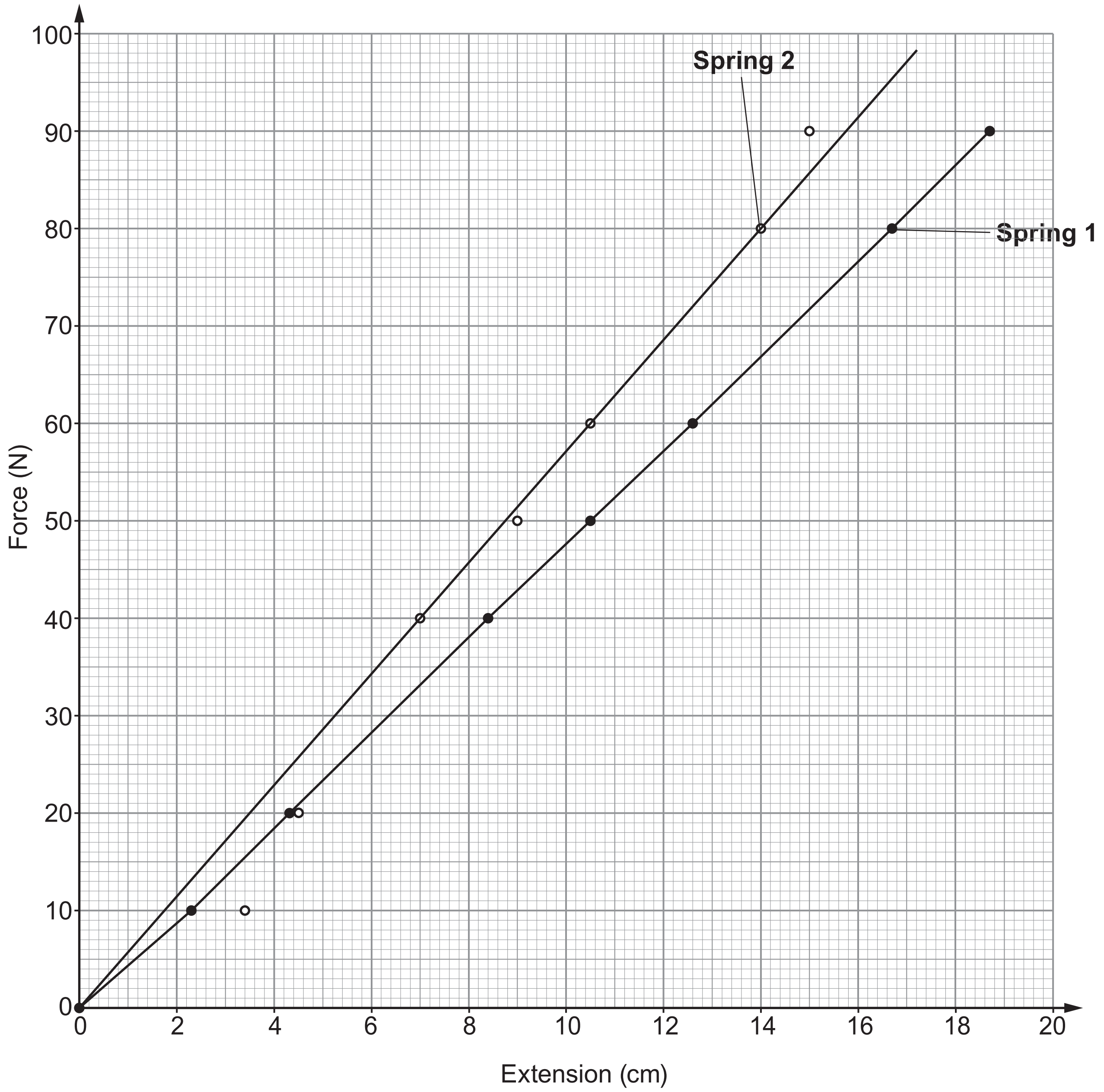
It is important that the spring is designed so that it does not become permanently stretched during the process. Springs become permanently stretched once they pass their **elastic limit**. This is the point where extension is no longer proportional to the stretching force.

Students test three springs using the apparatus shown below.



(a) Their results for two springs used in the doors are shown in the graph below.



Examiner only

(i) Compare the behaviour of the springs as the force increases from 10 N to 40 N. [2]

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(ii) Use the graph and the equation:

$$\text{spring constant} = \frac{\text{force}}{\text{extension}}$$

to calculate the value of the spring constant for **Spring 1**. [3]

spring constant = ... N/cm

(iii) Use the graph to state **one** reason why the students can have more confidence in the results for **Spring 1** than for **Spring 2**. [1]

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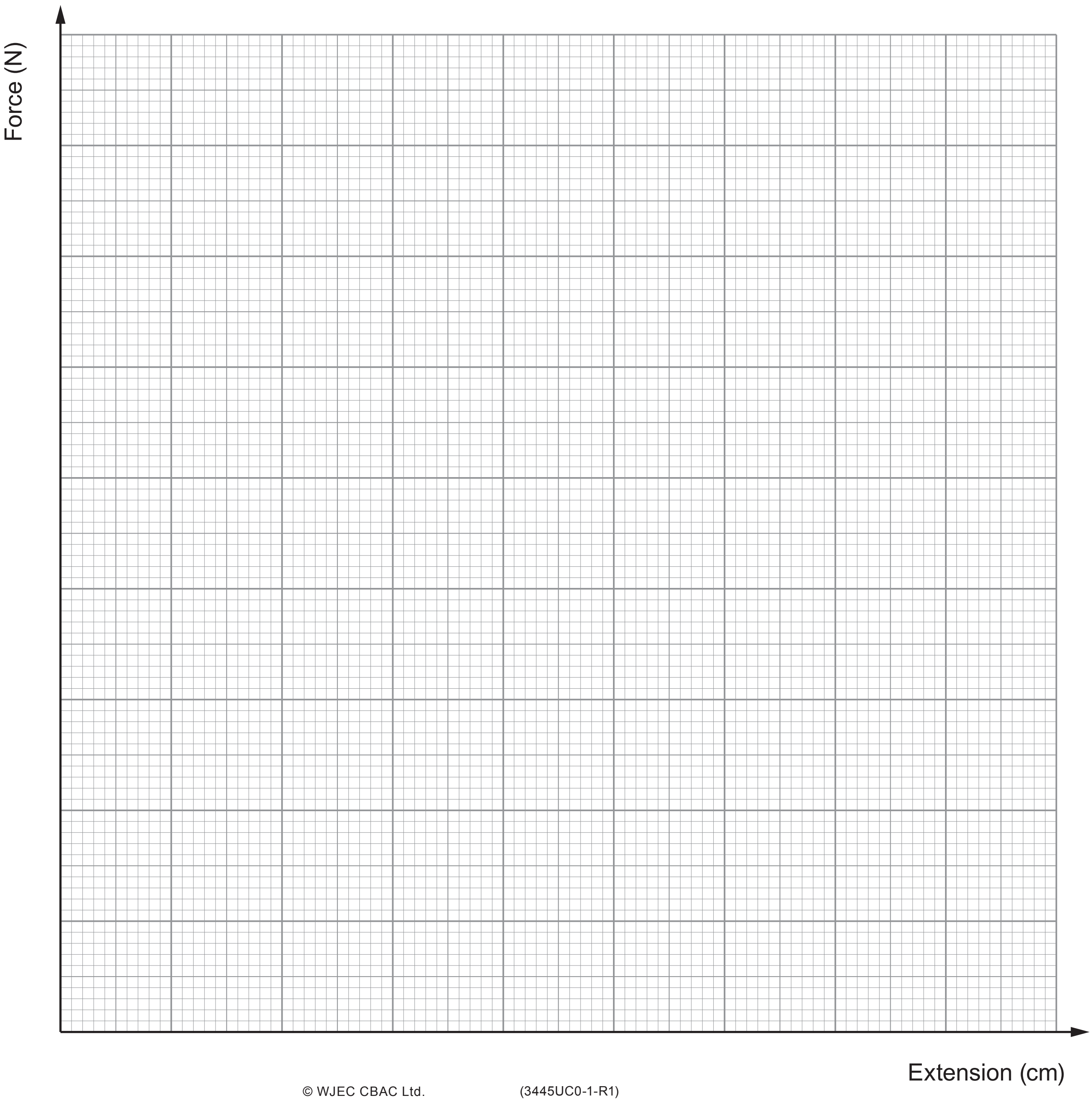
(b) The results for **Spring 3** are given in the table below.

Force (N)	Extension (cm)
0	0.0
10	2.4
20	4.8
40	9.6
50	12.0
60	16.0
80	28.0
90	36.0

(i) Plot the data on the grid below and draw a suitable line. [4]



Examiner only

(ii) Springs become permanently stretched once they pass their **elastic limit**. This is the point where extension is no longer proportional to the stretching force.

I. **Label** the elastic limit on your graph with a letter **E**. [1]

II. Use your graph to find the force required to reach the elastic limit. [1]

Force = ... N

(iii) During the opening and closing of the garage doors the spring will extend by 30 cm. Explain whether or not **Spring 3** is suitable for this purpose. [2]

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